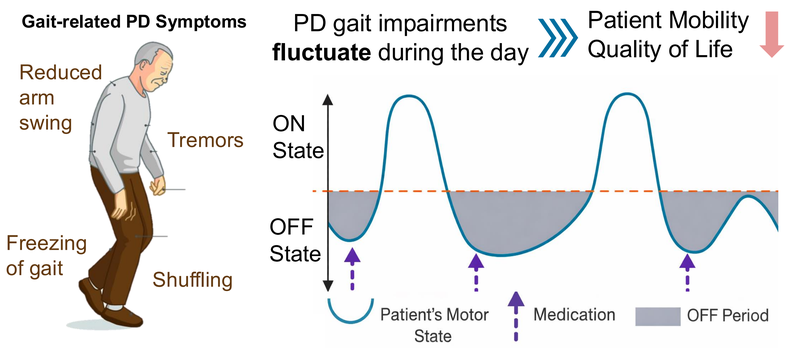
GaitWay: A Modular Wearable System for Continuous Gait Monitoring and Quantitative Assessment of Parkinson's Disease (PD)

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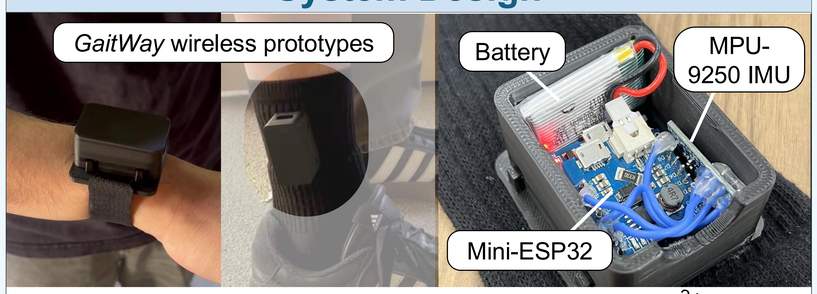
GaitWay

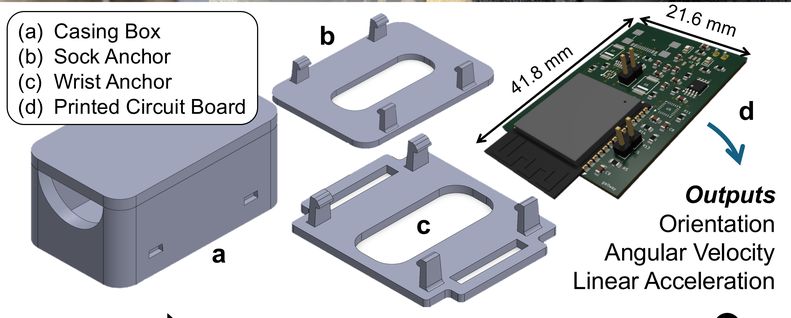
Background & Motivation



Lack of **affordable and wearable** continuous gait monitoring system

System Design



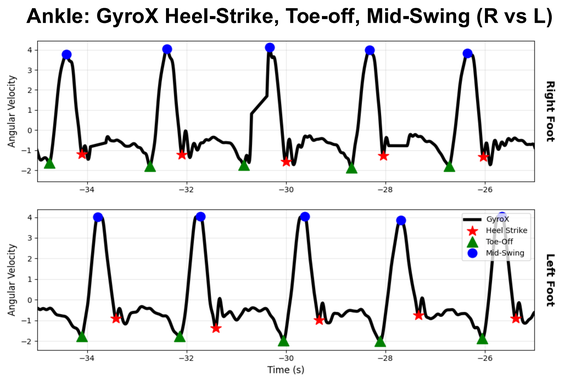


GaitWay sensor → BLE → Smartphone / Laptop → **Gait Score** → user

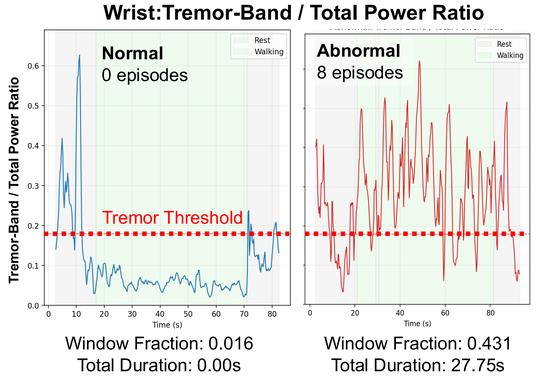
Experimental Design

Normal Walking	Abnormal Walking	1 foot Per Second	2 Feet Per Second	Increasing Stride Time
Self-selected pace / length	Shuffling with shorter steps	Metronome & Fixed distance		Changing metronome
Baseline	Deviation	Validation		Validation
R/L Asymmetry (%)		Stride Length (m)		Stride Duration (s)
Double Support (%)		Speed (m/s)		

Real-Time Data Acquisition and Gait Event Detection



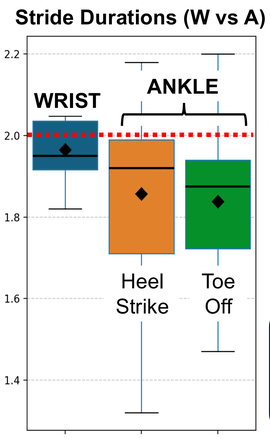
Ankle: Event Detection from Peak & Minima



Window Fraction: 0.016
Total Duration: 0.00s

Window Fraction: 0.431
Total Duration: 27.75s

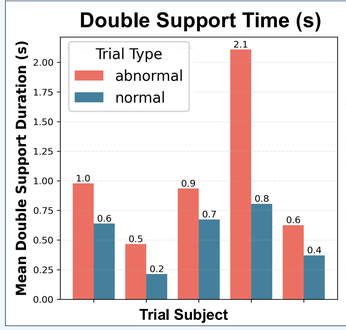
Wrist: Abnormal Tremor Band Analysis

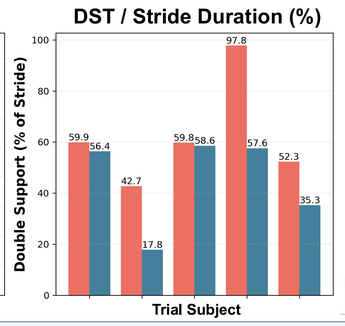




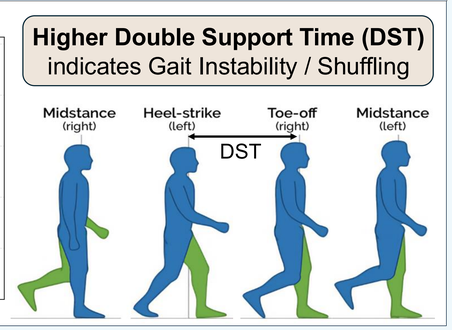
Decreasing / fixed metronome pacing
Increasing / constant stride duration over time

Parkinsonian Gait Symptoms - Normal vs Abnormal





Higher Double Support Time (DST) indicates Gait Instability / Shuffling



Machine-Learning Based Gait Score

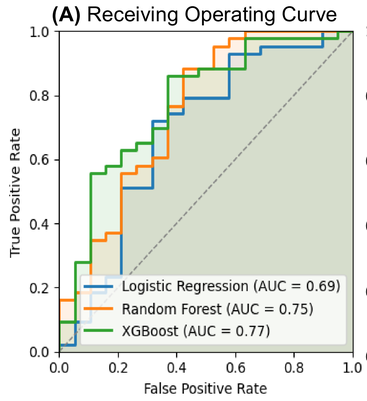
R/L Asymmetry* (%)

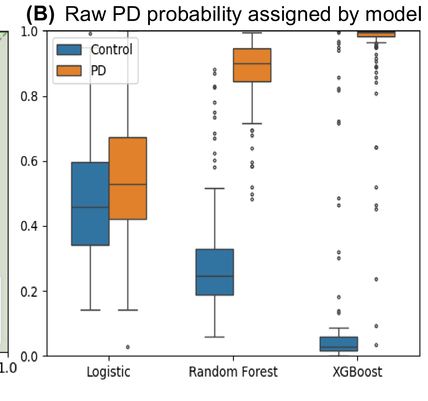
Stride Duration* (s) | Double-support* (%) | Stride Length* (m) | Speed* (m/s)

** Length of boxes depict relative feature importance*

IMU data → Lower Limb Parameters → *XGBoost Model* → Gait Characterization → **Gait Score**

IMU data → Upper Limb Parameters → *Algorithm* → Arm Swing Characterization → **Gait Score**



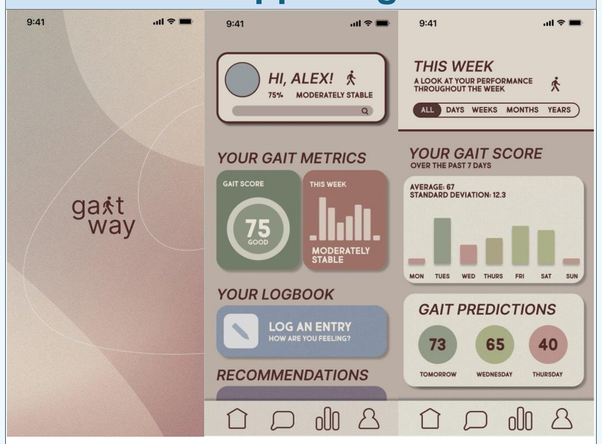


Chosen Model: XGBoost

(A) **Highest Area under ROC curve of 77%** (greatest probability of assigning higher score to PD patient vs healthy)

(B) **Clearest separation** in distribution of predicted PD probabilities for **PD patients** vs **Healthy**

Mobile App Integration



Display Gait Score, trends over time, & predictions
Log medications / subjective gait quality from patients

Future Directions

- Flexible PCB for improved wearability
- Vision-based tracking for validation
- Clinical trials with Parkinson's Disease patients
- Monitor medication & gait correlation

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